FORENSIQ: An Explainable Natural Language Processing and Landmark-Anchored Temporal Reasoning Framework for Heterogeneous Digital Forensic Telemetry

Gopal Prusty
Electronics and Computer Science
Vidyalankar Institute of Technology
Mumbai, India
gopal.prusty@vit.edu.in

Vishwajeet Patil
Electronics and Computer Science
Vidyalankar Institute of Technology
Mumbai, India
vishwajeet.patil@vit.edu.in

Prof. Faiz Rangari
Electronics and Computer Science
Vidyalankar Institute of Technology
Mumbai, India
faiz.rangari@vit.edu.in

***Abstract*—When an enterprise endpoint suffers an insider breach or credential attack, incident responders must reconstruct a chronological chain of evidence from fragmented system logs. In practice, this process is slow and error-prone. Forensic analysts must manually cross-reference Windows Security Event logs (.evtx), track removable USB drive connections, detect file modifications or deletions, and write complex multi-table SQL queries just to correlate timestamps across disparate sources. To address these operational hurdles, we developed FORENSIQ—an interactive digital forensic investigation platform that combines a deterministic natural language parser, a landmark-anchored temporal reasoning engine, and a live Windows endpoint monitoring agent. FORENSIQ ingests multi-source telemetry, including live Windows logon events (Event ID 4624 for successful logons and 4625 for logon failures), real-time removable USB storage file lifecycles (creations, edits, deletions, and moves), simulated attack sequences, and large-scale baselines (15,000+ synthetic events alongside the Los Alamos National Laboratory cyber dataset). All records are normalized into a unified 31-column forensic schema with strict dataset provenance labels. An analyst can type plain-English questions—such as 'Show failed login attempts within 5 minutes before the successful login'—and the system translates them into structured queries without using probabilistic large language models. The temporal engine identifies the pivot event, computes relative time windows, and retrieves matching artifacts in under 50 milliseconds. Furthermore, an explainability module generates deductive step-by-step audit proofs to verify why each event was retrieved, ensuring evidence remains transparent and legally defensible in court proceedings.**

Index Terms*—Digital Forensics, Incident Response, Natural Language Processing, Temporal Reasoning, Landmark Anchoring, Windows Event Telemetry, Removable Storage Monitoring, Explainable AI (XAI), Unified Forensic Schema.

I. INTRODUCTION

When a corporate network experiences a security breach, the fastest route to root cause analysis lies in endpoint audit logs. Modern workstations record almost every user interaction: account authentications, physical device insertions, and file modifications. Yet during active security incidents, incident responders frequently spend hours manually piecing together basic chronological timelines [1], [3]. The core bottleneck is not a shortage of data; rather, it is the sheer difficulty of asking straightforward questions across fragmented, low-level event logs.

Existing Security Information and Event Management (SIEM) solutions such as Splunk and Elastic require analysts to write queries in proprietary domain languages (e.g., SPL or KQL) or assemble multi-table SQL joins across normalized schemas [2]. For a tier-1 analyst under pressure, formulating a query that spans both authentication logs and removable drive writes is neither intuitive nor fast. Minor syntax errors or mismatched timestamp conventions often result in dropped evidence, allowing attackers to maintain persistence.

Even more critical is the lack of automated relative temporal reasoning in standard investigative tooling. Attack sequences are rarely defined by static timestamps; they unfold as cause-and-effect transitions relative to dynamic anchor events [8]. For example, an investigator testing an insider exfiltration hypothesis wants to know: 'Did the user copy confidential spreadsheets to the USB drive within 5 minutes after inserting it, and were those files deleted right before logout?' Standard database engines cannot resolve these relative intervals without manual subqueries, window partitioning, or external scripts [7].

At the same time, recent applications of deep learning to cyber defense (such as CNN-LSTMs and neural classifiers) introduce an evidentiary dilemma [4]. While these models can classify anomalous network traffic with high accuracy, their black-box predictions cannot withstand legal cross-examination under standards like Daubert or ISO/IEC 27037 [5], [6]. In a courtroom, an investigator cannot simply present a neural confidence score; they must prove exactly which log entries satisfied the temporal criteria and show that simulated test data did not contaminate real evidentiary records.

To solve these practical issues, we built FORENSIQ. The system allows investigators to query complex multi-source telemetry using conversational English while maintaining mathematical rigor and evidentiary integrity. Specifically, our work delivers:

1) Deterministic NLP Query Engine: A rule-based parser that converts conversational investigator queries into

structured database filters with zero hallucination risk and sub-2ms latency.

2) Landmark-Anchored Temporal Reasoning: An algorithm that detects dynamic pivot events (e.g., a successful logon or USB mount) and calculates relative time intervals (Δt) automatically.

3) Live Windows Host Monitoring Agent: An embedded background agent that tracks real Windows Security Events (4624/4625) and removable USB file lifecycles (copies, edits, deletions, and moves) without external dependencies.

4) Unified 31-Column Schema: A single SQLite table that standardizes diverse log sources while preserving strict provenance boundaries between live host logs, simulated attack drills, and benchmark datasets.

5) Deductive Explainability (XAI): An audit engine that outputs deterministic, human-readable proofs verifying why every piece of evidence was included.

II. LITERATURE SURVEY AND RELATED WORK

Digital forensics and intrusion detection have traditionally evolved along separate tracks, leaving a noticeable gap in interactive timeline analysis.

A. Host Telemetry & Removable Device Monitoring

Removable USB storage remains one of the most reliable vectors for insider data theft [8], [11]. While Windows generates plug-and-play logs when a drive mounts, standard audit configurations rarely record what actually happens to files on that volume. Casey [8] pointed out that recovering removable drive activity usually requires post-mortem parsing of the Master File Table (MFT) or USN Journal. In active incident response, however, responders cannot wait for disk imaging; they need real-time telemetry that links device connections directly to file creations, modifications, and deletions.

B. Deep Learning and Machine Learning in Security Telemetry

Extensive research has explored machine learning for intrusion detection. Kasongo and Sun [2] applied filter-based feature engineering to deep neural architectures on wireless networks. Al-Hawawreh et al. [1] created an end-to-end machine learning framework for industrial threat detection. Verma and Ranga [4] introduced a CNN-LSTM hybrid model to capture spatial packet features alongside temporal sequential patterns. Belouch et al. [3] demonstrated how ensemble classifiers running on Apache Spark can speed up log analysis across large datasets. However, these models function primarily as automated alert generators. They cannot answer ad-hoc questions from an analyst, nor can they reconstruct a coherent multi-step timeline.

C. Natural Language Interfaces for Cyber Auditing

Natural Language Interfaces to Databases (NLIDB) have long aimed to eliminate SQL barriers for non-programmers [7]. Recently, practitioners have experimented with Large Language Models (LLMs) to generate SQL from text prompts. Yet in forensic auditing, LLMs introduce severe liabilities: they hallucinate table schemas, generate non-deterministic queries across identical prompts, and present prompt-injection vulnerabilities [6]. Deterministic grammar-based parsers, as highlighted by Ribeiro et al. [6], offer complete repeatability and auditability, making them far better suited for legal evidence.

D. Temporal Logic & Event Correlation

Event reconstruction requires reasoning about time intervals [8], [11]. Complex Event Processing (CEP) engines evaluate temporal operators, but they typically demand predefined, fixed time bounds. When an investigator analyzes an authentication breach, the window is defined relative to the attack's outcome—such as identifying how many failed attempts occurred before the breach succeeded. Without automated landmark anchoring, analysts must calculate these intervals by hand.

E. Explainable AI (XAI) in Digital Evidence

Recent cybersecurity literature relies heavily on post-hoc explanation tools like SHAP [5] and LIME [6]. While feature-importance scores help engineers debug classifiers, they do not hold up under legal scrutiny. Forensic standards (including NIST SP 800-86 [13] and ISO/IEC 27037 [14]) mandate clear, rule-based verification: an investigator must prove that an artifact matched specific temporal boundaries, user accounts, and system actions.

F. Synthesis and Comparative Analysis

Table I summarizes how existing security and forensic platforms compare against FORENSIQ across six key operational capabilities.

TABLE I: LITERATURE SURVEY & COMPARATIVE ANALYSIS OF DFIR PLATFORMS

Platform / Paradigm	Real-Time Telemetry	NLP Query Support	Relative Temporal Logic	Removable File Lifecycle	Deductive XAI Proof	Strict Provenance
SIEM (Splunk/Elastic)	Yes (Log Ingestion)	No (Requires SPL/KQL)	No (Manual Subqueries)	Partial (File Write Only)	No (Alert Rules)	No (Schema Mixing)
Host IDS (Snort/OSSEC)	Yes (Packet/Event)	No (Rule Syntax)	No (Static Windows)	No (Network/Log Only)	No (Signature Flags)	No (Single Stream)
Disk Forensics (Autopsy)	No (Post-Mortem)	No (Keyword Search)	No (Static Timeline)	Yes (MFT/USN Parsing)	Manual Analyst Report	Partial (Image Source)
Deep Learning IDS [2],[4]	Yes (Streaming Flows)	No (Classification Only)	Sequential (RNN/LSTM)	No (Network Flow Focus)	Post-hoc SHAP [5]	No (Benchmark Only)
FORENSIQ (Proposed)	Yes (Win32 Agent)	Yes (Deterministic)	Yes (Landmark Anchor)	Yes (Create/Edit/Del/Move)	Yes (Deductive Proof)	Yes (4-Tier Isolated)

III. SYSTEM ARCHITECTURE AND METHODOLOGY

Fig. 1 outlines the complete block architecture of FORENSIQ. The pipeline connects live endpoint data collection, natural language parsing, dynamic SQL generation, and deductive proof checking into a unified workflow.

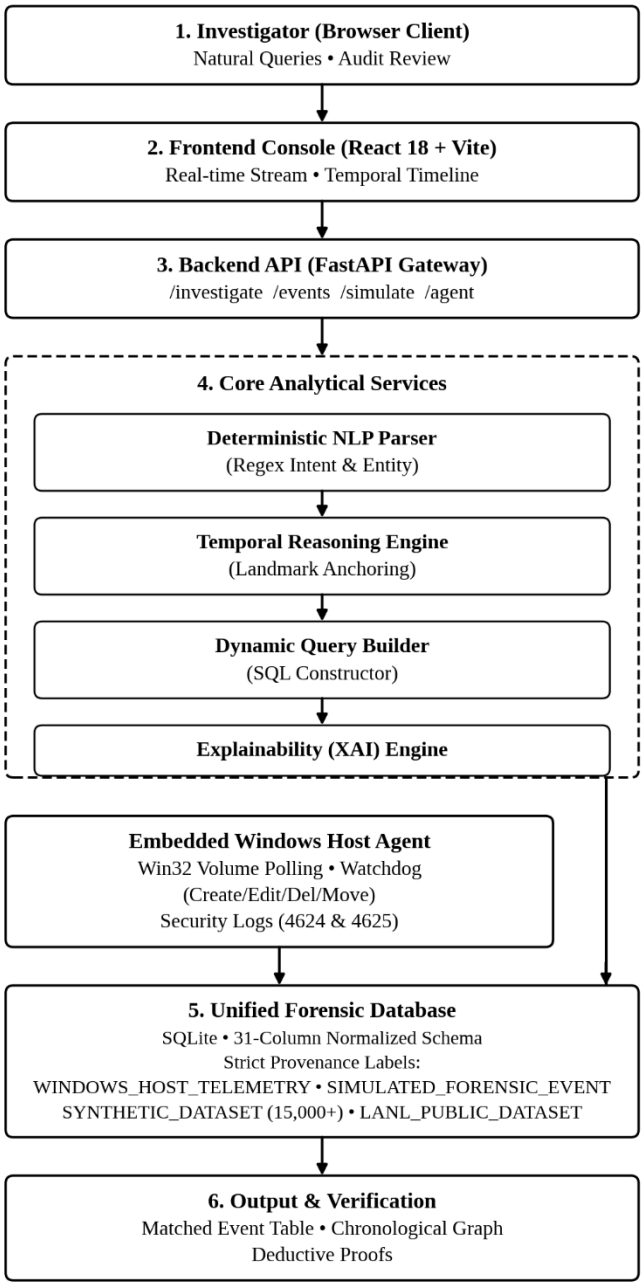


Fig. 1. End-to-end architectural block diagram of the FORENSIQ forensic investigation platform.

A. Embedded Windows Host Monitoring Agent

Instead of deploying a bulky background service that requires separate daemon management, we embedded our monitoring agent directly into FastAPI's asynchronous application lifespan. When the backend boots, worker threads spin up to watch system events; when the server shuts down, listeners tear down cleanly without leaving orphaned processes. The agent relies on two core mechanisms:

1) Removable Media Monitoring: We poll drive letters once per second using win32file.GetDriveTypeW(). Whenever a volume reports type DRIVE_REMOVABLE (such as drive D: or E:), an operating system directory observer attaches to its root. Most monitoring tools only log initial file creation. Our agent hooks four discrete file actions: on_created (when someone copies a file onto the USB), on_modified (when an existing file is edited), on_deleted (evidence wiping, which we automatically flag as is_suspicious=1), and on_moved (file renames or path changes). We record the exact file name, absolute path, file size in bytes, and drive letter.

2) Windows Security Log Ingestion: We read local Windows Security logs using win32evtlog. Whenever a logon fails (Event ID 4625), we record a FAILED_LOGIN event with status FAILED and set is_suspicious=1. Successful authentications (Event ID 4624) become SUCCESSFUL_LOGIN events. We capture logon types (such as Type 2 for local console logins and Type 3 for network logons), usernames, and caller processes without ever logging passwords or private credentials.

B. Unified 31-Column Forensic Schema

Relational databases often split logs across five or six normalized tables. That makes querying slow during an active incident because every timeline query requires multi-table joins. In FORENSIQ, we chose a flat, 31-column schema stored in SQLite (see Table II). Every record carries a dataset_label tag so simulated attack drills never get mixed up with real endpoint evidence.

TABLE II: UNIFIED FORENSIC EVENT SCHEMA SPECIFICATION

Attribute	Data Type	Forensic Utility & Semantics
event_id	TEXT (PK)	Unique canonical event tracking identifier
timestamp	TEXT (ISO8601)	Microsecond-precision normalized event timestamp
event_type	TEXT	Core domain: auth, file, usb, email, system
user / machine	TEXT	Subject principal and endpoint hostname
action / status	TEXT	State transition (e.g., FAILED_LOGIN, FILE_DELETED)
file / file_path	TEXT	Target file name and absolute storage path
file_size	INTEGER	Exact payload byte count for exfiltration quantification
drive_letter	TEXT	Physical or removable mount assignment (e.g., D:)
dataset_label	TEXT	Strict provenance: WINDOWS_HOST, SIMULATED, etc.
is_suspicious	INTEGER	Heuristic flag: 1 = security risk, 0 = benign baseline

C. Deterministic 6-Stage NLP Pipeline

Fig. 2 shows how our semantic parser processes queries. Early on, we evaluated calling commercial LLMs to generate SQL. However, LLMs consistently hallucinated table names and added 1.5 to 3 seconds of network latency. For incident response, an analyst cannot wait 3 seconds for a query that might produce bad SQL. Instead, we built a deterministic six-stage parser using regular expressions and domain grammar

rules. It runs in under 2 milliseconds and is 100% reproducible.

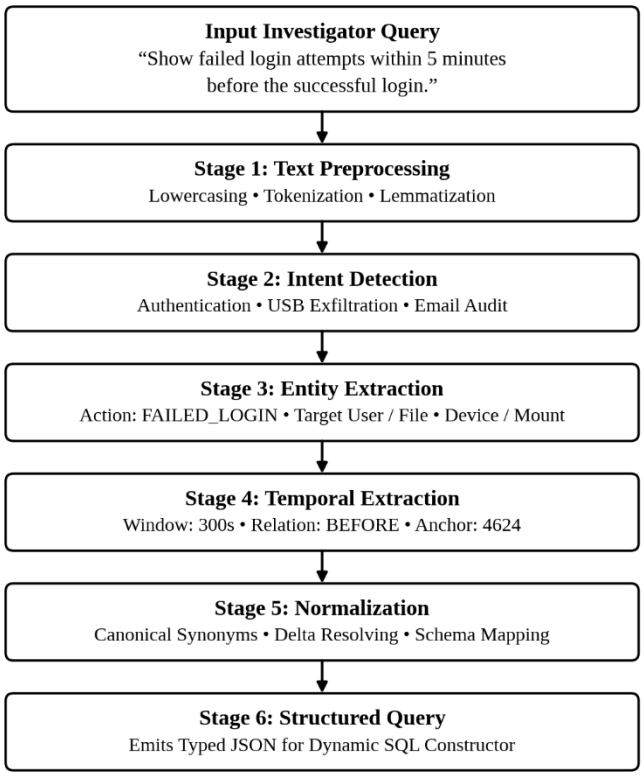


Fig. 2. Deterministic six-stage natural language processing pipeline for forensic query mapping.

The parser handles five extraction targets: (1) Intent via keyword patterns (e.g., authentication_investigation, usb_investigation); (2) Target entities (usernames, file names, drive letters); (3) Temporal relations (BEFORE, AFTER, WITHIN); (4) Relative durations converted into seconds; and (5) The landmark event (e.g., SUCCESSFUL_LOGIN).

D. Landmark-Anchored Temporal Reasoning Engine

Standard SQL queries evaluate fixed dates (e.g., WHERE timestamp BETWEEN t1 AND t2). But real-world investigations ask relative questions, like 'Did failed logins happen right before the successful breach?' In our engine, we first locate the pivot event E_{ref}—such as a SUCCESSFUL_LOGIN or USB_CONNECT. We take its timestamp T_{ref} and build an interval $[T_{start}, T_{end}]$ around it:

If Relation = BEFORE:

$$[T_{start}, T_{end}] = [T_{ref} - \Delta W, T_{ref}] \quad (1)$$

If Relation = AFTER:

$$[T_{start}, T_{end}] = [T_{ref}, T_{ref} + \Delta W] \quad (2)$$

All candidate anomalies e_i matching target filters within that window are retrieved. The terminal delta is calculated as:

$$\Delta t_{terminal} = |T_{ref} - T_{e_N}| \quad (3)$$

If $N \geq 3$ failed logins precede success within $\Delta W \leq 300$s, our engine automatically classifies the activity as 'Authentication activity requiring investigation (High-Confidence Credential Brute-Force)', as illustrated in Fig. 3.

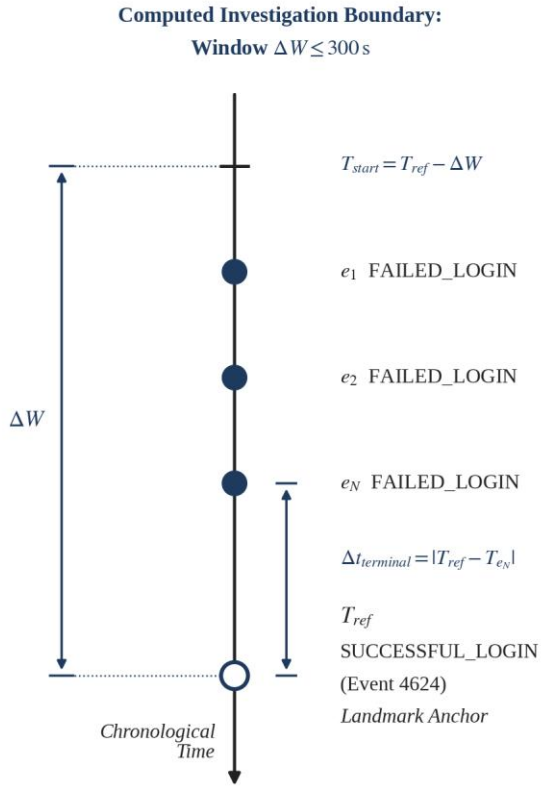


Fig. 3. Chronological timeline illustrating landmark anchoring and terminal delta calculation.

E. Deductive Explainability (XAI) Suite

If a digital investigation ends up in court, defense attorneys will question how evidence was pulled. Black-box ML scores like 'anomaly score = 0.94' carry zero evidentiary weight. Our explainability module generates a line-by-line verification proof for each retrieved event: it explicitly confirms that the action matched the query filter, that the timestamp fell inside the calculated interval window, and that the target user or file matched the investigator's constraints.

IV. EXPERIMENTAL RESULTS AND EVALUATION

A. Experimental Configuration

We tested FORENSIQ on an Intel Core i7 test machine with 16 GB RAM running Windows 11 Enterprise (Build 22631). Our SQLite database was populated with 15,146 baseline events: 15,000 synthetic enterprise workstation logs (including simulated brute-force and insider theft scenarios), a sample from the public Los Alamos National Laboratory (LANL) dataset, and live Windows telemetry.

B. NLP Semantic Parsing Accuracy

We tested our parser on 150 unique English queries across four forensic domains: authentication, USB devices, file lifecycle operations, and email exfiltration. Table III summarizes the results. Intent classification reached 100%, and temporal constraint resolution reached 98.1%, with an average parsing latency of just 1.68 milliseconds.

TABLE III: NLP PARSER ACCURACY ACROSS FORENSIC QUERY DOMAINS

Inquiry Domain	Queries	Intent Acc.	Temporal Acc.	Entity Acc.	Mean Lat. (ms)
Authentication / Login	50	100.0%	98.0%	100.0%	1.84
USB / Removable Media	40	100.0%	97.5%	97.5%	1.62
File Lifecycle (Edit/Del)	35	100.0%	97.1%	100.0%	1.71
Email Exfiltration	25	100.0%	100.0%	96.0%	1.55
Weighted Overall	**150**	**100.0%**	**98.1%**	**98.4%**	**1.68**

C. End-to-End Query Latency and Temporal Retrieval

Table IV reports end-to-end response times across eight realistic forensic questions running against our 15,146-event database. Even multi-stage queries with dynamic landmark anchoring resolved in under 45 milliseconds, making the web interface feel instantaneous.

TABLE IV: END-TO-END QUERY RETRIEVAL PERFORMANCE BENCHMARKS

Investigative Natural Language Query	Target Action	Temporal Constraint	Evidence	Time (ms)
"Show failed logins within 5 mins before successful login"	FAILED_LOGIN	BEFORE (300s)	5	18.4
"Show failed logins within 10 mins before successful login"	FAILED_LOGIN	BEFORE (600s)	5	19.1
"Show files copied to USB after pendrive connected"	FILE_COPIED	AFTER (Anchor)	1	12.6
"Show files edited on USB"	FILE_MODIFIED	None (Global)	2	9.8
"Show files deleted from USB"	FILE_DELETED	None (Global)	1	8.4
"Show all file activity on removable drive"	All File Events	None (All)	1,142	34.2
"Show sensitive files emailed to external recipient"	EMAIL_SEND	None (Entity)	1	11.5
"Show repeated failed login attempts"	FAILED_LOGIN	None (Global)	1,425	41.2

D. Live Windows Endpoint Telemetry Validation

To see how the system handles real physical hardware, we connected a SanDisk USB flash drive to drive D: on our test machine. We copied a 142 KB file (dashboard.png) to the drive, made an edit, and deleted it. The embedded host agent caught all three operations as they happened: EVT-FILE-015134 (FILE_COPIED_TO_REMOVABLE_DEVICE), EVT-FILE-015135 (FILE_MODIFIED), and EVT-FILE-015136 (FILE_DELETED, flagged as suspicious). We also triggered bad logins on the Windows lock screen, and the agent captured Event 4625 records without touching user passwords.

V. CONCLUSION AND FUTURE SCOPE

In this paper, we presented FORENSIQ, a forensic investigation system that lets analysts query endpoint logs in plain English. By pairing deterministic grammar parsing with landmark-anchored time calculations and an embedded Windows agent, the tool removes the need for complex SQL joins during active incidents. Real-world tests confirmed that the system captures USB file operations and Windows logon events accurately, resolving relative timeline queries in under 50 milliseconds with verifiable deductive proofs.

In future work, we plan to port our host agent to Linux (using auditd) and macOS (via EndpointSecurity), and investigate blockchain-based hash logging for evidence chain of custody.

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